Redox Reactions & Equivalent Concept

Position in the Periodic Table & Transition Bridge (NCERT §8.1)

The d-block comprises elements of Groups 3 to 12 occupying Periods 4, 5, 6, and 7. They bridge the highly electropositive, ionic s-block metals on the left and the electronegative, covalent p-block non-metals on the right, exhibiting intermediate transitional character.

1. IUPAC Definition & The Group 12 Enigma

- **IUPAC Formal Definition:** A transition element is defined as an element which has an **incompletely filled d-subshell (d^1 to d^9)** in its ground state OR in any of its common oxidation states.
- **Group 12 Elements (Zn, Cd, Hg):** Zinc ([Ar] $3d^{10}$ $4s^2$), Cadmium ([Kr] $4d^{10}$ $5s^2$), and Mercury ([Xe] $4f^{14}$ $5d^{10}$ $6s^2$) have completely filled $(n-1)d^{10}$ subshells in both their elemental ground state and their common +2 oxidation states (Zn^{2+}, Cd^{2+}, Hg^{2+}).
- **Why Studied with Transition Metals?** They conclude the d-series, display continuous physical trends (complexation with Lewis bases, soft acid behavior), but strictly lack quintessential transition properties: they are diamagnetic, form colorless compounds, and have unusually low melting points (Hg is liquid at room temperature!).
- **Four Transition Series:** 3d series (Sc 21 to Zn 30 in Period 4), 4d series (Y 39 to Cd 48 in Period 5), 5d series (La 57, Hf 72 to Hg 80 in Period 6, interrupted by 14 4f lanthanoids), and 6d series (Ac 89, Rf 104 to Cn 112 in Period 7, interrupted by 14 5f actinoids).

2. Valence Configurations & Origin of Anomalies

The general valence configuration is **$(n-1)d^{1-10}$ ns^{1-2}**. In neutral atoms, Aufbau principle places 4s below 3d ($E_{4s} < E_{3d}$). However, upon ionization, **4s electrons are ALWAYS lost first** because once 3d orbitals are populated, increased nuclear shielding causes 3d to sink deeper in energy than 4s ($E_{3d} < E_{4s}$ in cations!).

Element	Atomic No.	Expected Config.	Observed Config.	Physical Driving Force
Chromium (Cr)	24	[Ar] $3d^4$ $4s^2$	**[Ar] $3d^5$ $4s^1$**	Exchange energy maximization: 10 exchange pairs vs 6 pairs + spherical symmetry
Copper (Cu)	29	[Ar] $3d^9$ $4s^2$	**[Ar] $3d^{10}$ $4s^1$**	Extraordinary stability of completely filled d^{10} subshell with minimal repulsions
Palladium (Pd)	46	[Kr] $4d^8$ $5s^2$	**[Kr] $4d^{10}$ $5s^0$**	Completely filled $4d^{10}$; total absence of valence 5s electrons!
Platinum (Pt)	78	[Xe] $4f^{14}$ $5d^8$ $6s^2$	**[Xe] $4f^{14}$ $5d^9$ $6s^1$**	Relativistic stabilization of 6s and d-shell exchange energy
Gold (Au)	79	[Xe] $4f^{14}$ $5d^9$ $6s^2$	**[Xe] $4f^{14}$ $5d^{10}$ $6s^1$**	Filled $5d^{10}$ + strong relativistic contraction of valence 6s orbital

Table 1.1 — Anomalous ground-state electronic configurations in d-block.

TRAP

CRITICAL TRAP: Writing Fe^{2+} configuration as [Ar] $3d^4$ $4s^2$ is completely WRONG! In all transition cations, 4s electrons are lost FIRST: Fe^{2+} is [Ar] $3d^6$ $4s^0$, Sc^{3+} is [Ar] $3d^0$, and Cu^+ is [Ar] $3d^{10}$.

Lattice Cohesion & The Dual Metallic-Covalent Bonding Model (NCERT §8.2)

All transition elements are typical metals possessing high tensile strength, ductility, malleability, high thermal/electrical conductivities, and metallic luster. They crystallize in dense close-packed structures: bcc, ccp (fcc), and hcp lattices.

1. Dual Nature of Bonding in Transition Metals

- **Metallic Sea + Covalent d-d Overlap:** In transition metals, bonding involves BOTH delocalized valence **ns electrons** (typical metallic bonding) AND directional overlap of **unpaired (n−1)d electrons** between adjacent atoms (covalent bonding).
- **Hardness & Boiling Points:** Greater the number of unpaired d-electrons, stronger is the covalent interatomic bonding → metals are exceptionally hard with low volatility.
- **Group 12 Softness:** Zn, Cd, and Hg have no unpaired d electrons (d^{10}). Covalent d-d overlap is absent; metallic bonding is weak → soft metals with low boiling points; Hg is a liquid at 298 K ($\Delta_{vap} H^\circ$ = 59 kJ/mol).

2. Melting Points & The Anomalous Dip at Mn and Tc

Melting points rise across a series, peaking at **Group 6 (Cr, Mo, W)** with maximum unpaired electrons, then decrease regularly towards Group 12. **Tungsten (W, 3422 °C)** holds the highest melting point of all metals, followed by Re and Os.

- **The Mn (1246 °C) and Tc Dip:** Despite having 5 unpaired d-electrons, Manganese and Technetium display anomalously low melting points!
- **Reason:** Symmetrical, tightly held half-filled **$3d^5$ (and $4d^5$)** configuration holds electrons firmly to the nucleus, minimizing their delocalization into the metallic lattice. Furthermore, Mn adopts a complex distorted cubic lattice with unusually long interatomic distances.

3. Enthalpies of Atomization ($\Delta_a H^\circ$) (NCERT §8.3.1)

Enthalpy of atomization is the energy required to dissociate 1 mole of solid metal into gaseous atoms: M(s) → M(g). It is a direct quantitative metric of metallic lattice strength.

Series	Peak Element	Peak $\Delta_a H^\circ$ (kJ/mol)	Group 12 Min. (kJ/mol)	Physical Origin & Trend
3d Series	Vanadium (V)	515 kJ/mol	Zn (126 kJ/mol)	Unpaired 3d electrons maximize lattice cohesion; Zn has d^{10}
4d Series	Niobium / Mo	726 / 659 kJ/mol	Cd (112 kJ/mol)	Larger 4d radial extent enhances orbital overlap
5d Series	Tungsten (W)	**849 kJ/mol**	Hg (61 kJ/mol)	**5d > 4d > 3d**: Diffuse 5d orbitals yield powerful M-M bonding!

Table 2.1 — Atomization enthalpies across transition series.

TRAP
EXAM TRAP: Group 6 metals have maximum atomization energies, but 5d series metals have systematically HIGHER atomization enthalpies than 3d and 4d congeners. This explains why 5d elements are chemically noble, hard, and form stable metal-metal cluster compounds (e.g. $[Re_2Cl_8]^{2-}$, W_6Cl_{12}).

Effective Nuclear Charge vs Screening Effect (NCERT §8.3.2)

Atomic and ionic sizes dictate the chemical reactivity, hydration energetics, and coordination geometries of transition metal complexes. Across each transition series, radii exhibit a unique tripartite profile rather than a monotonic decline.

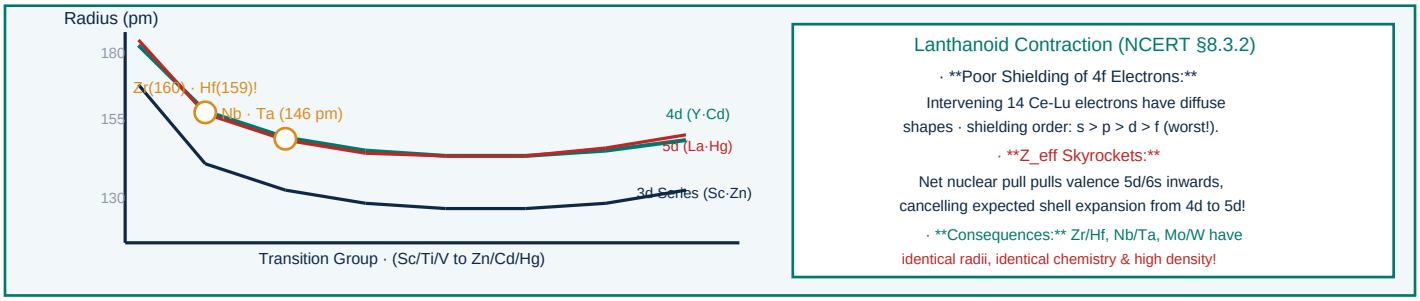


Fig. 3.1 — Atomic radii of 3d, 4d, and 5d transition series illustrating Lanthanoid Contraction.

1. Atomic Radii Profile across 3d Series (Sc to Zn)

- **1. Sc to Cr (Steep Decrease):** Added electrons enter inner 3d subshell. Increase in nuclear charge (Z_{eff}) heavily outweighs the poor mutual shielding provided by d-electrons → valence cloud contracts (Sc: 162 pm → Cr: 128 pm).
- **2. Mn to Ni (Nearly Flat Plateau):** Added 3d electrons now provide increased screening that virtually balances the increment in nuclear charge ($Z_{eff} \approx \text{screening}$) → size remains nearly constant (Fe: 126 pm, Co: 125 pm, Ni: 125 pm).
- **3. Cu to Zn (Slight Expansion):** Filling of the completely paired $3d^{10}$ subshell creates strong inter-electronic repulsions that overpower nuclear attraction, causing electron cloud expansion (Cu: 128 pm → Zn: 137 pm).

2. Lanthanoid Contraction: The 4d ≈ 5d Chemical Twin Phenomenon

Down a periodic group, atomic size normally increases due to addition of extra electron shells (Sc: 162 pm → Y: 180 pm). However, between 4d and 5d series, **radii are virtually IDENTICAL** (Zr: 160 pm ≈ Hf: 159 pm; Nb: 146 pm ≈ Ta: 146 pm; Mo: 139 pm ≈ W: 139 pm)!

- **Physical Cause:** The intervening filling of **14 4f electrons** before 5d series (Cerium to Lutetium). 4f orbitals are diffuse with extremely poor shielding capability (screening order: s > p > d > f).
- **Nuclear Dominance:** Nuclear charge increases by +32 units from 4d to 5d, but poorly screened 4f electrons allow this enormous nuclear charge to pull valence 5d and 6s shells inward with massive force, completely neutralizing expected shell expansion.
- **Major Consequences:** (1) Zr/Hf and Nb/Ta are chemical twins with identical chemical behavior; (2) Densities of 5d metals double (Os = 22.59 g/cm³, Ir = 22.56 g/cm³); (3) Basic strength of hydroxides decreases steadily from $La(OH)_3$ to $Lu(OH)_3$.

TRAP
JEE ADVANCED HIGHLIGHT: Due to Lanthanoid Contraction, Hf has a HIGHER ionization enthalpy than Zr, and separation of Zr and Hf requires fractional distillation of volatile alkoxides or ion-exchange chromatography; chemical precipitation cannot separate them!

Ionization Energetics across 3d, 4d, and 5d Series (NCERT §8.3.3)

Ionization enthalpy increases across each transition series, but the rate of increase is far less steep than in s and p blocks because incoming electrons enter an inner (n−1)d subshell, partially shielding the outer ns electrons from nuclear pull.

1. Analysis of First, Second, and Third Ionization Enthalpies

- **First Ionization Enthalpy (IE_1):** Ranges from Sc (631 kJ/mol) to Zn (906 kJ/mol). Zn exhibits the highest IE_1 because the electron must be stripped from a stable, completely filled $4s^2$ orbital.
- **IE_2 Anomalies (Chromium and Copper):**
- - **Cr (1592 kJ/mol):** Second electron is removed from stable half-filled **$3d^5$** (Cr^+: [Ar] $3d^5$ → Cr^{2+}: $3d^4$). Loss of huge exchange energy requires massive energy.
- - **Cu (1958 kJ/mol):** Second electron is removed from exceptionally stable completely filled **$3d^{10}$** (Cu^+: [Ar] $3d^{10}$ → Cu^{2+}: $3d^9$).
- **IE_3 Anomalies (Manganese, Iron, and Zinc):**
- - **Mn (3248 kJ/mol):** Third electron must be stripped from stable half-filled **$3d^5$** (Mn^{2+}: [Ar] $3d^5$ → Mn^{3+}: $3d^4$). Hence, Mn^{2+} resists oxidation to Mn^{3+}.
- - **Fe (2957 kJ/mol):** Relatively low IE_3 because removing third electron yields stable **$3d^5$** (Fe^{2+}: [Ar] $3d^6$ → Fe^{3+}: $3d^5$). Thus, Fe^{2+} is easily oxidized to Fe^{3+}.
- - **Zn (3833 kJ/mol):** Requires disruption of completely filled, stable $3d^{10}$ core (Zn^{2+}: [Ar] $3d^{10}$ → Zn^{3+}: $3d^9$).

2. Down the Group: The Anomalous 5d > 3d > 4d Hierarchy

In s and p blocks, IE decreases down a group due to increased atomic radius. In transition groups, IE follows the order: **5d > 3d > 4d** (or 5d > 4d > 3d in several groups).

- **Why 5d has the HIGHEST Ionization Enthalpy:** Because of **Lanthanoid Contraction**, 5d elements experience an enormous increase in nuclear charge without size increase. Valence 6s electrons are held tightly by the nucleus.
- **Chemical Inertness of Heavy Metals:** Exceptionally high IE values of 5d metals (Pt, Au, Hg) explain why Gold and Platinum are chemically noble, resisting dissolution in standard mineral acids.

3. Thermodynamic Balance: Cu+ vs Cu2+ in Aqueous Media

Thermodynamic stability in water depends on net energy balance ($\Delta_{\text{sub}}H + \text{IE}_1 + \text{IE}_2 + \Delta_{\text{hyd}}H$): · Although IE_2 of Cu is high (+1958 kJ/mol), **$Cu^{2+}(aq)$ is far more stable than $Cu^+(aq)$** in water because high charge density of Cu^{2+} yields an immense negative hydration enthalpy ($\Delta_{hyd}H^\circ = -2121$ kJ/mol), which more than compensates for IE_2! · In aqueous solution, cuprous ion undergoes spontaneous disproportionation: **$2\,Cu^+(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$** ($E^\circ = +0.37\,\text{V}$).

TRAP
EXAM TRAP: Cu^+ is stable in insoluble solids (CuCl, CuBr, CuI) and complexes with π-acceptor ligands ($[Cu(CN)_2]^-$), but rapidly disproportionates in water!

Electronic Flexibility & High-Oxidation State Thermodynamics (NCERT §8.3.4)

Unlike s-block metals which exhibit fixed oxidation states (+1 or +2), transition metals display a wide spectrum of oxidation states due to the very small energy gap between (n−1)d and ns orbitals, allowing both to participate in chemical bonding.

1. Oxidation State Spectrum in the 3d Series

Element	Outer Config.	All Observed States	Most Stable State	Dominant Chemical Character
Sc	$3d^1\ 4s^2$	+3	**+3**	d^0 noble core; diamagnetic, colorless
Ti	$3d^2\ 4s^2$	+2, +3, +4	**+4**	d^0 covalent/ionic; TiO_2, $TiCl_4$
V	$3d^3\ 4s^2$	+2, +3, +4, +5	**+4, +5**	+5 is oxidizing (V_2O_5); +4 is stable $[VO]^{2+}$
Cr	$3d^5\ 4s^1$	+2, +3, +4, +5, +6	**+3, +6**	+3 (t_{2g}^3 CFSE stability); +6 is strong oxidant
Mn	$3d^5\ 4s^2$	+2, +3, +4, +5, +6, +7	**+2, +4, +7**	Maximum states (+2 to +7); +2 (d^5) is stable
Fe	$3d^6\ 4s^2$	+2, +3, +4, +6	**+2, +3**	+3 (d^5 stable); ferrate $[FeO_4]^{2-}$ in alkali
Co	$3d^7\ 4s^2$	+2, +3, +4	**+2, +3**	+2 in simple salts; +3 in strong field complexes
Ni	$3d^8\ 4s^2$	+2, +3, +4	**+2**	+2 is thermodynamically dominant (Ni^{2+})
Cu	$3d^{10}\ 4s^1$	+1, +2	**+2 (aq)**	+1 in solid halides; +2 in aqueous solution
Zn	$3d^{10}\ 4s^2$	+2	**+2**	Only +2; completely filled $3d^{10}$ shell

Table 5.1 — Oxidation states of 3d transition metals.

2. Group Trends: Contrast Between d-Block and p-Block

- **p-Block vs d-Block Contrast:** In p-block (Groups 14, 15), lower oxidation states become more stable down the group due to the inert pair effect (Pb^{2+} > Pb^{4+}, Bi^{3+} > Bi^{5+}).
- **In d-Block: HIGHER oxidation states become MORE STABLE down the group!**
- - **Group 6 Proof:** Cr(+6) in dichromate $[Cr_2O_7]^{2-}$ is a **strong oxidizing agent**, whereas Mo(+6) in MoO_3 and W(+6) in WO_3 are **extraordinarily stable and non-oxidizing**!
- - **Group 8 Proof:** Fe tops out at +6 in unstable ferrate, while Ru and Os easily form stable volatile tetroxides: **RuO_4 and OsO_4 (+8 state!)**.
- **Physical Reason:** Greater radial extension of 4d and 5d orbitals allows effective multi-center covalent overlap with electronegative ligands, stabilizing higher formal oxidation states.

TRAP
CRITICAL JEE ADVANCED TRAP: Acidic nature increases with increasing oxidation state: MnO (basic) < Mn_3O_4 < Mn_2O_3 < MnO_2 (amphoteric) < Mn_2O_7 (strongly acidic covalent green oil!).

Thermodynamic Anatomy of Reduction Potentials (NCERT §8.3.5)

The standard electrode potential $E^{\circ}(M^{2+}/M)$ measures the free energy change for the reduction: $M^{2+}(aq) + 2e^- \rightarrow M(s)$. It represents the net thermodynamic competition between sublimation, ionization, and hydration enthalpies.

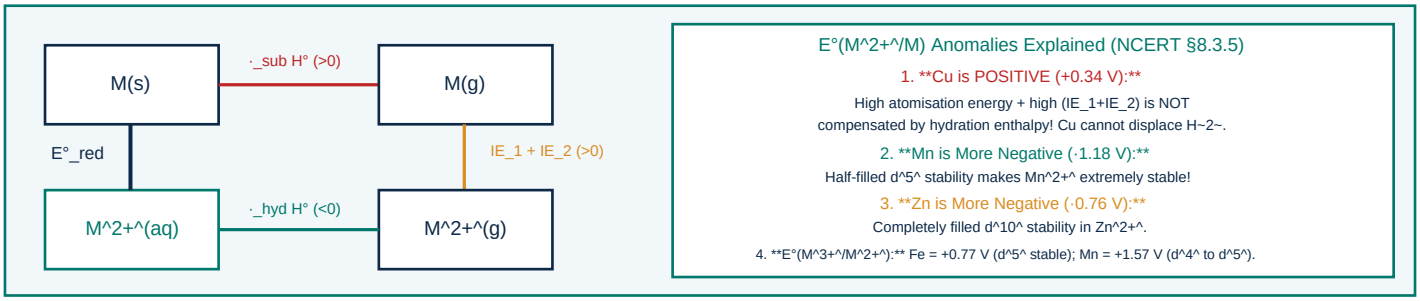


Fig. 6.1 — Thermochemical Born-Haber cycle for $E^{\circ}(M_2+/M)$ and 3d redox irregularities.

1. Analysis of E°(M2+/M) Irregularities across 3d Series

- **General Trend:** Values become less negative across the series, reflecting the general rise in (IE_1 + IE_2).
- **The Copper Anomaly ($E^{\circ} = +0.34\text{ V}$):** Copper is the ONLY 3d metal with a **positive reduction potential**! Reason: High enthalpy of atomization (+339 kJ/mol) combined with high (IE_1 + IE_2 = +2703 kJ/mol) is NOT compensated by its hydration enthalpy (−2121 kJ/mol). Cu cannot displace H_2 from non-oxidizing mineral acids (HCl, dil H_2SO_4).
- **Manganese Peak ($E^{\circ} = -1.18\text{ V}$):** More negative than expected because Mn^{2+} attains extra exchange stability of **half-filled $3d^5$** shell, lowering IE_2.
- **Zinc Peak ($E^{\circ} = -0.76\text{ V}$):** More negative due to the extra stability of completely filled **$3d^{10}$** in Zn^{2+}.
- **Nickel Peak ($E^{\circ} = -0.25\text{ V}$):** More negative than predicted due to an exceptionally large negative hydration enthalpy.

2. The E°(M3+/M2+) Redox Couple System

- **Manganese ($E^{\circ} = +1.57\text{ V}$):** Extremely positive! Mn^{3+} ($3d^4$) desperately seeks an electron to become stable $3d^5$ (Mn^{2+}). Hence, Mn^{3+} is a powerful oxidizing agent.
- **Chromium ($E^{\circ} = -0.41\text{ V}$):** Negative value! Cr^{2+} ($3d^4$) is a **strong reducing agent** because oxidation yields Cr^{3+} with half-filled stable **t_{2g}^3** configuration in octahedral crystal field (CFSE stabilization!).
- **Iron ($E^{\circ} = +0.77\text{ V}$):** Moderate positive value. Fe^{3+} has stable $3d^5$, making Fe^{2+} easily oxidizable by air to Fe^{3+}.

TRAP
EXAM TRAP: Cr^{2+} and Mn^{3+} both have $3d^4$ configuration, yet Cr^{2+} is reducing while Mn^{3+} is oxidizing! Always justify via t_{2g}^3 crystal field stability for Cr^{3+} vs $3d^5$ exchange stability for Mn^{2+}.

Ligand Stabilization of Extreme Valency (NCERT §8.3.6)

The highest oxidation states in transition metals are stabilized exclusively by the most electronegative elements: **Fluorine and Oxygen**. However, their stabilization mechanisms differ fundamentally in bonding geometry and electron capacity.

1. Transition Metal Halides & Maximum Oxidation States

- **Fluorine Superiority in Halides:** Only fluorides can stabilize maximum oxidation states in halides due to small ionic radius, high electronegativity, and massive lattice enthalpy (or high M-F bond energy). E.g. **TiF_4 (+4), VF_5 (+5), CrF_6 (+6)**.
- **Manganese Halide Barrier:** The highest manganese halide is **MnF_4 (+4)**; Mn(+7) cannot be formed with fluorine as MnF_7. However, Mn(+7) is readily stabilized with oxygen as **Mn_2O_7** or oxohalide **MnO_3F**.
- **FeI_3 and CuI_2 Non-Existence:**
- - **FeI_3 does not exist!** Fe^{3+} oxidizes iodide to iodine: $2\ Fe^{3+} + 2\ I^- \rightarrow 2\ Fe^{2+} + I_2$.
- - **CuI_2 does not exist!** Cu^{2+} oxidizes iodide to iodine: $2\ Cu^{2+} + 4\ I^- \rightarrow 2\ CuI\downarrow (white) + I_2$. CuI (cuprous iodide) is precipitated.

2. Oxides vs Fluorides: Oxygen's Unique Advantage

Although fluorine has higher electronegativity than oxygen, **oxygen stabilizes higher oxidation states better than fluorine!**

- **Physical Mechanism:** Oxygen has the unique ability to form **multiple pπ-dπ bonds** with the transition metal using its 2p lone pairs into vacant metal d-orbitals.
- **Case Study — Manganese:** Highest fluoride is MnF_4 (+4), but highest oxide is **Mn_2O_7 (+7)**, a green explosive covalent oil where two tetrahedral $[MnO_4]$ units share a bridging oxygen (Mn-O-Mn).
- **Case Study — Vanadium & Chromium:** Vanadium forms stable oxo-cations: **$[VO_2]^+$ (+5)**, **$[VO]^{2+}$ (+4)**; Chromium forms **CrO_3 (+6)** and **CrO_2F_2** (chromyl fluoride), stabilized by terminal oxo double bonds.

3. Acidity and Covalency Gradient of 3d Oxides

Metal & Ox. State	Formula	Acid-Base Nature	Bonding Character	Aqueous Reaction / Species
Cr(+2)	CrO	**Basic**	Predominantly Ionic	Dissolves in acids to give blue $Cr^{2+}(aq)$
Cr(+3)	Cr_2O_3	**Amphoteric**	Intermediate Covalent	Soluble in both mineral acids and conc. NaOH
Cr(+6)	CrO_3	**Strongly Acidic**	Covalent Network	Anhydride of H_2CrO_4; dissolves to give $[CrO_4]^{2-}$
Mn(+2)	MnO	**Basic**	Ionic	Dissolves in acid to yield pink $Mn^{2+}(aq)$
Mn(+7)	Mn_2O_7	**Strongly Acidic**	Covalent Oil	Explosive oil; anhydride of permanganic acid $HMnO_4$

Table 7.1 — Acid-base and bonding trends of chromium and manganese oxides.

TRAP
EXAM TRAP: V_2O_5 is amphoteric: dissolves in acids to give dioxovanadium(V) cation $[VO_2]^+$ and in strong bases to give orthovanadate anion $[VO_4]^{3-}$.

Origin of Magnetism in Coordination & Solid State Systems (NCERT §8.3.7)

Magnetic moments in transition metal complexes originate from two distinct physical phenomena: (1) electron spin angular momentum (S), and (2) electron orbital angular momentum (L).

1. Quenching of Orbital Angular Momentum in 3d Metals

- **The Quenching Phenomenon:** In 3d transition metals, outer 3d orbitals extend towards the periphery of the atom and interact strongly with the electrostatic crystal field generated by surrounding ligands/ions. This strong electric field quenches (cancels) the orbital angular momentum ($L \approx 0$).
- **Spin-Only Formula (Bohr Magnetons, BM):**
- $$\mu_s = \sqrt{n(n+2)} \text{ BM} = 2\sqrt{S(S+1)} \text{ BM}$$
- where n is the number of unpaired d-electrons, $S = n/2$ is the total spin, and $1\text{ BM} = \frac{eh}{4\pi m_e} = 9.274 \times 10^{-24} \text{ J/T}$.

2. Direct Value Quick-Lookup Table

Unpaired e^- (n)	Spin (S)	Formula $\sqrt{n(n+2)}$	Calc. μ_s (BM)	Experimental Range (BM)	Typical 3d Ions
1	1/2	$\sqrt{3}$	**1.73 BM**	1.75 – 1.90	Ti^{3+} ($3d^1$), Cu^{2+} ($3d^9$), $[VO]^{2+}$ ($3d^1$)
2	1	$\sqrt{8}$	**2.83 BM**	2.80 – 2.90	Ti^{2+} ($3d^2$), V^{3+} ($3d^2$), Ni^{2+} ($3d^8$)
3	3/2	$\sqrt{15}$	**3.87 BM**	3.80 – 3.95	V^{2+} ($3d^3$), Cr^{3+} ($3d^3$), Co^{2+} ($3d^7$)
4	2	$\sqrt{24}$	**4.90 BM**	4.80 – 5.00	Cr^{2+} ($3d^4$), Fe^{2+} ($3d^6$), Mn^{3+} ($3d^4$)
5	5/2	$\sqrt{35}$	**5.92 BM**	5.85 – 6.00	Mn^{2+} ($3d^5$), Fe^{3+} ($3d^5$) — Maximum in 3d!

Table 8.1 — Unpaired electrons vs calculated spin-only magnetic moments.

3. Solid-State Magnetism & 4d/5d/Lanthanoid Deviations

- **Diamagnetism:** All electrons paired ($n = 0$). Repelled by magnetic field. E.g. Sc^{3+} (d^0), Ti^{4+} (d^0), Zn^{2+} (d^{10}), Cu^+ (d^{10}).
- **Paramagnetism:** Unpaired electrons present ($n \ge 1$). Attracted into magnetic field; obeys Curie's Law ($\chi = C/T$).
- **Ferromagnetism:** Permanent alignment of magnetic domains (Fe, Co, Ni, CrO_2). CrO_2 is used in magnetic recording tapes.
- **Why Spin-Only Formula FAILS in 4d, 5d & Lanthanoids:** Heavy elements possess strong spin-orbit coupling; orbital angular momentum is NOT quenched. Total angular momentum J must be used: $\mu = g_J \sqrt{J(J+1)}$. Exception: **Gd^{3+} ($4f^7$, L = 0)** obeys the spin-only formula perfectly ($n = 7$, $\mu = 7.94\text{ BM}$)!

TRAP

CRITICAL TRAP: When calculating magnetic moments, always check for strong field vs weak field ligands in octahedral coordination: $[Fe(CN)_6]^{3-}$ has $n = 1$ (low spin, $\mu = 1.73\text{ BM}$), whereas $[Fe(H_2O)_6]^{3+}$ has $n = 5$ (high spin, $\mu = 5.92\text{ BM}$).

Redox Reactions & Equivalent Concept

Electronic Spectroscopy of Transition Ions (NCERT §8.3.8)
Most transition metal compounds exhibit vivid colors in solid state and aqueous solutions. Color arises when visible light is absorbed to promote an electron between split d-orbitals (d-d transition) or between ligand and metal orbitals (charge transfer).

1. Mechanism of Colour: d-d Electronic Transitions

- **Crystal Field Splitting:** In an isolated gaseous atom, all 5 d-orbitals are degenerate. When ligands (like H_2O) apprpach in octahedral geometry, orbitals split into lower energy t_{2g} (d_{xy}, d_{yz}, d_{zx}) and higher energy e_g ($d_{x^2-y^2}$, d_{z^2}) levels.
- **Complementary Absorption:** Absorption of a photon matching Δ_o ($h\nu$) excites an electron from t_{2g} to e_g. The transmitted light displays the **complementary color**.
- **Why d^{0} and d^{10} Ions are Colourless:**
- **d^0 Ions:** Sc^{3+}, Ti^{4+}, V^{5+} have NO d-electrons to promote → colorless.
- **d^{10} Ions:** Cu^{+}, Zn^{2+}, Cd^{2+}, Hg^{2+} have no vacant d-orbitals to receive promoted electrons → colorless.

2. Charge Transfer (CT): Intensely Coloured d0 Systems

The Grand Paradox: Why are $[MnO_4]^-$ (intense purple), $[Cr_2O_7]^{2-}$ (bright orange), and $[CrO_4]^{2-}$ (vivid yellow) intensely coloured even though Mn(+7) and Cr(+6) are **d^0 systems**?

- **Mechanism: Ligand-to-Metal Charge Transfer (LMCT):**
- Because the central metal is in a very high oxidation state with strong positive pull, an electron temporarily migrates from the ligand (oxide O^{2-} $2p$ orbital) to the metal's empty d-orbitals upon absorption of visible light.
- **Laporte Rule & Intensity:** LMCT transitions are fully Laporte-allowed (p to d involves parity change $g \rightarrow u$), yielding molar extinction coefficients $\sim 10^{4}\ \text{L mol}^{-1}\text{ cm}^{-1}$ (1000x more intense than weak Laporte-forbidden d-d transitions!).
- **Other Famous CT Systems:** $Fe(SCN)_3$ (blood red), HgI_2 (scarlet red), AgI (yellow).

3. Standard Reference Table of Aqueous 3d Ion Colours

Ion	Valence Config.	Unpaired e^-	Hydrated Aqueous Colour	Spectroscopic Mechanism
Ti^{3+}	$3d^1$	1	**Purple / Violet**	d-d transition ($t_{2g}^1\ e_g^0 \rightarrow t_{2g}^0\ e_g^1$)
$[VO]^{2+}$	$3d^1$	1	**Blue**	d-d transition in oxovanadium(IV)
Cr^{3+}	$3d^3$	3	**Violet-Green**	d-d transition (t_{2g}^3 configuration)
Cr^{2+}	$3d^4$	4	**Sky Blue**	d-d transition
Mn^{2+}	$3d^5$	5	**Very Pale Pink**	Spin-forbidden d-d transition (very weak intensity)
Fe^{3+}	$3d^5$	5	**Yellow / Amber**	Spin-forbidden d-d + slight LMCT from water
Fe^{2+}	$3d^6$	4	**Pale Green**	d-d transition
Co^{2+}	$3d^7$	3	**Pink (Oct.) / Deep Blue (Tet.)**	$[Co(H_2O)_6]^{2+}$ is pink; $[CoCl_4]^{2-}$ is deep blue
Ni^{2+}	$3d^8$	2	**Bright Green**	d-d transition in $[Ni(H_2O)_6]^{2+}$
Cu^{2+}	$3d^9$	1	**Blue**	d-d transition in distorted octahedral $[Cu(H_2O)_6]^{2+}$

Table 9.1 — Standard colors of 3d transition metal ions in aqueous solution.

TRAP
EXAM TRAP: Anhydrous $CuSO_4$ is pure WHITE because in the absence of water ligands, there is no crystal field splitting ($\Delta = 0$). Only hydrated $CuSO_4 \cdot 5H_2O$ is blue!

Coordination Capacity & Advanced Materials (NCERT §8.3.9 - §8.3.12)

Transition metals dominate coordination chemistry and heterogeneous catalysis due to their unique electronic and structural adaptability.

1. Four Factors Driving Coordination Complex Formation

- **1. Small Ionic Size & High Nuclear Charge:** Generates extraordinarily high charge-to-radius ratio (charge density), pulling ligand electron pairs strongly.
- **2. Vacant Low-Energy Orbitals:** Availability of vacant (n–1)d, ns, and np orbitals allows acceptance of lone pairs donated by Lewis bases.
- **3. Variable Oxidation States:** Enables flexible geometric rearrangements to stabilize diverse coordination numbers (4, 6).
- **4. Crystal Field Stabilization Energy (CFSE):** Splitting of d-orbitals yields substantial thermodynamic CFSE gain.

2. Catalytic Properties & Dual Mechanisms

- **Mechanism A: Variable Valency (Homogeneous Catalysis):** Transition metal ions cycle between oxidation states to provide a lower activation energy pathway. E.g. Persulfate-iodide reaction catalyzed by Fe^{3+}:
- - Step 1: $2\ Fe^{\{3+\}} + 2\ I_ \rightarrow 2\ Fe\{2+\} + I_2$
- - Step 2: $2\ Fe^{\{2+\}} + S_2O_8\{2-\} \rightarrow 2\ Fe^{\{3+\}} + 2\ SO_4\{2-\}$
- **Mechanism B: Adsorption & Chemisorption (Heterogeneous Catalysis):** Utilizing vacant d-orbitals to form temporary chemisorption bonds with gaseous reactants, increasing surface concentration and weakening intramolecular bonds.
- **Industrial Catalysts:** Fe in Haber's Process (NH_3); V_2O_5 in Contact Process ($SO_2 \rightarrow SO_3$); Finely divided Ni in hydrogenation; $TiCl_4 + Al(C_2H_5)_3$ (Ziegler-Natta Catalyst) in ethene polymerization.

3. Interstitial Compounds & Alloy Formation

- **Interstitial Compounds:** Formed when small non-metal atoms (H, C, N, B) become trapped in interstitial voids of metal lattices (e.g. TiC, Fe_3H, Mn_4N, $VH_{0.56}$).
- - **Characteristics:** Non-stoichiometric; very high melting points (higher than pure metals); extremely hard (TiC approaches diamond hardness); retain metallic conductivity; chemically inert.
- **Alloy Formation:** Transition metals have atomic radii within 15% of each other (Hume-Rothery rules), allowing atoms to substitute mutually in crystal lattices.
- - **Brass:** Cu (60-80%) + Zn (20-40%); **Bronze:** Cu (80%) + Sn (20%); **Stainless Steel:** Fe + Cr (18%) + Ni (8%) + C.

TRAP
JEE ADVANCED HIGHLIGHT: Interstitial hydrides (e.g. $VH_{0.56}$) are non-stoichiometric and do not follow standard valence rules; their electrical conductivities are comparable to parent metals!

Synthesis from Chromite Ore & Structural Interconversion (NCERT §8.4)

Potassium dichromate is an indispensable laboratory primary standard and industrial oxidizing agent manufactured from chromite ore (ferrochrome, $FeCr_2O_4$).

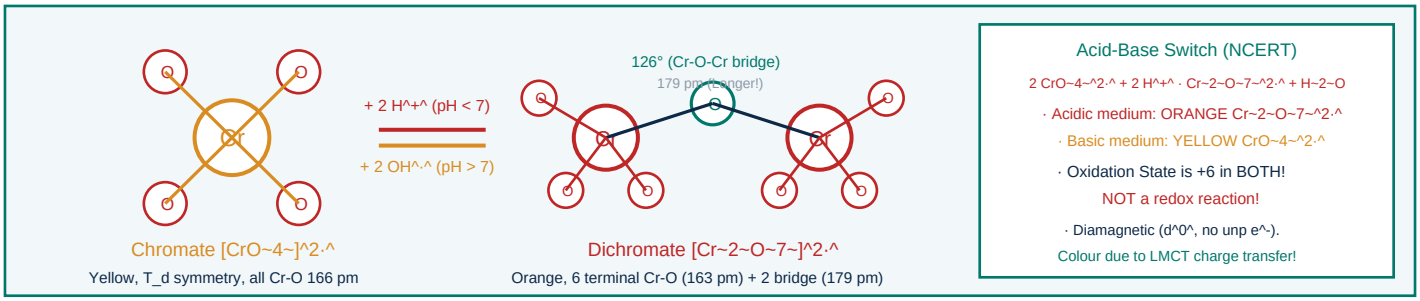


Fig. II.1 — Tetrahedral chromate vs bridged dichromate and the acid-base equilibrium.

1. Three-Step Industrial Manufacture

- **Step 1: Oxidative Alkaline Roasting with Soda Ash:**
- Powdered chromite ore is fused with sodium carbonate and quicklime in excess air in a reverberatory furnace:
- $4 FeCr_2O_4 + 8 Na_2CO_3 + 7 O_2 \rightarrow 8 Na_2CrO_4 (yellow) + 2 Fe_2O_3 (brown\downarrow) + 8 CO_2$
- Roasted mass is leached with water: soluble yellow Na_2CrO_4 dissolves, leaving insoluble Fe_2O_3 behind.
- **Step 2: Acidification to Sodium Dichromate:**
- Yellow chromate solution is filtered and acidified with concentrated H_2SO_4:
- $2 Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7 (orange) + Na_2SO_4\downarrow + H_2O$
- Less soluble Na_2SO_4 crystallizes out as Glauber's salt ($Na_2SO_4 \cdot 10H_2O$) upon cooling and is removed.
- **Step 3: Conversion to Potassium Dichromate:**
- Hot concentrated sodium dichromate is treated with potassium chloride (KCl):
- $Na_2Cr_2O_7 + 2 KCl \rightarrow K_2Cr_2O_7 (orange\downarrow) + 2 NaCl$
- $K_2Cr_2O_7$ is much less soluble in cold water than NaCl and crystallizes out as bright orange-red crystals.

2. The Chromate-Dichromate pH Switch

- **Dynamic Reversible Equilibrium:**
- $2 CrO_4^{2-} (yellow) + 2 H^+ \rightleftharpoons Cr_2O_7^{2-} (orange) + H_2O$
- - **In Acidic Medium (pH < 7):** Adding H^+ drives equilibrium to the right → **Orange $Cr_2O_7^{2-}$** dominates.
- - **In Basic Medium (pH > 7):** Adding OH^- consumes H^+ → equilibrium shifts to the left → **Yellow CrO_4^{2-}** dominates.
- **Oxidation State is +6 in BOTH ions!** This is an acid-base condensation reaction, **NOT a redox process.**
- **Structural Comparison:**
- - **Chromate ($[CrO_4]^{2-}$):** Tetrahedral (T_d symmetry); all four Cr-O bonds are identical (**166 pm**) due to resonance.
- - **Dichromate ($[Cr_2O_7]^{2-}$):** Two tetrahedra sharing a common oxo-bridge. Cr-O-Cr angle is **126°**. Bridging Cr-O bonds are longer (**179 pm**); six terminal Cr-O bonds are shorter (**163 pm**).

> TRAP
> WHY $K_2Cr_2O_7$ AND NOT $Na_2Cr_2O_7$ IN TITRATIONS? $K_2Cr_2O_7$ is non-hygroscopic and can be weighed directly as a primary volumetric standard. $Na_2Cr_2O_7$ is highly deliquescent (absorbs atmospheric moisture), making accurate weighing impossible!

Volumetric Oxidant & Confirmatory Qualitative Tests (NCERT §8.4)

In acidic solution, potassium dichromate acts as a powerful oxidizing agent. Its equivalent weight is strictly governed by a 6-electron transfer.

1. Acidic Redox Reactions & Equivalent Weight

In presence of dilute H_2SO_4, dichromate is reduced to green Cr^{3+}: $$Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \rightarrow 2\,Cr^{3+}\ (\text{green}) + 7\,H_2O \quad (E^\circ = +1.33\ \text{V}) \qquad \mathbf{\text{Equivalent Weight}} = \frac{\text{Molar Mass}}{6} = \frac{294.18}{6} = 49.03\ \text{g/eq}$$

Reducing Agent	Oxidized Product	Ionic Equation	Analytical Indicator
Fe^{2+} (Ferrous)	Fe^{3+} (Ferric)	$Cr_2O_7^{2-} + 14\,H^+ + 6\,Fe^{2+} \rightarrow 2\,Cr^{3+} + 6\,Fe^{3+} + 7\,H_2O$	Diphenylamine (internal)
I^- (Iodide)	I_2 (Iodine)	$Cr_2O_7^{2-} + 14\,H^+ + 6\,I^- \rightarrow 2\,Cr^{3+} + 3\,I_2 + 7\,H_2O$	Starch indicator (titrated with hypo)
H_2S (Hydrogen sulfide)	S (Colloidal sulfur)	$Cr_2O_7^{2-} + 8\,H^+ + 3\,H_2S \rightarrow 2\,Cr^{3+} + 3\,S\downarrow\ (\text{white}) + 7\,H_2O$	Turbid white colloidal sulfur
SO_3^{2-} (Sulfite)	SO_4^{2-} (Sulfate)	$Cr_2O_7^{2-} + 8\,H^+ + 3\,SO_3^{2-} \rightarrow 2\,Cr^{3+} + 3\,SO_4^{2-} + 4\,H_2O$	Turns orange solution green
Sn^{2+} (Stannous)	Sn^{4+} (Stannic)	$Cr_2O_7^{2-} + 14\,H^+ + 3\,Sn^{2+} \rightarrow 2\,Cr^{3+} + 3\,Sn^{4+} + 7\,H_2O$	Redox titration endpoint

Table 12.1 — Quantitative oxidation reactions of $K_2Cr_2O_7$ in acidic medium.

2. Chromyl Chloride Test (Confirmatory Test for Cl⁻)

- **Reaction:** *Solid chloride is heated with solid $K_2Cr_2O_7$ and concentrated H_2SO_4:*
- $$K_2Cr_2O_7 + 4\,KCl + 6\,H_2SO_4 \rightarrow 2\,CrO_2Cl_2\uparrow\ (\text{deep red vapor}) + 6\,KHSO_4 + 3\,H_2O$$
- **NaOH Absorption:** *Deep red vapors of chromyl chloride (CrO_2Cl_2) are absorbed in NaOH solution forming yellow sodium chromate:* $CrO_2Cl_2 + 4\,NaOH \rightarrow Na_2CrO_4\ (\text{yellow}) + 2\,NaCl + 2\,H_2O$.
- **Confirmation:** *Acidifying with acetic acid and adding lead acetate gives a* ***bright yellow precipitate of lead chromate***: $Na_2CrO_4 + Pb(CH_3COO)_2 \rightarrow PbCrO_4\downarrow\ (\text{yellow}) + 2\,CH_3COONa$.
- **JEE Exceptions:** *Chlorides of heavy metals (**Hg_2Cl_2, $HgCl_2$, $AgCl$, $PbCl_2$, $SnCl_2$**) DO NOT give this test because they are covalent or sparingly soluble. Bromides and iodides give Br_2 and I_2, NOT chromyl halides.*

3. Chromium Pentoxide (CrO_5): The Blue Butterfly Peroxide

When an acidified dichromate solution is treated with H_2O_2 in presence of ether: $$Cr_2O_7^{2-} + 2\,H^+ + 4\,H_2O_2 \rightarrow 2\,CrO_5\ (\text{deep blue}) + 5\,H_2O$$ • **Structure:** Butterfly structure with ONE oxo (*Cr=O*) and TWO peroxo (*O-O*) linkages. • **Oxidation State:** Oxo is −2; each peroxo oxygen is −1. Net: $x + (-2) + 4(-1) = 0 \implies \mathbf{x = +6}$!

> **TRAP**
> *EXAM TRAP: The deep blue color of CrO_5 fades rapidly in water as it decomposes to green Cr^{3+} and O_2; ether stabilizes it by coordinating to the vacant axial site of chromium!*

Manufacture from Pyrolusite Ore & Structural Properties (NCERT §8.4)

Potassium permanganate ($KMnO_4$) is the most widely utilized transition metal oxidizing agent in analytical and synthetic chemistry, manufactured from pyrolusite ore (MnO_2).

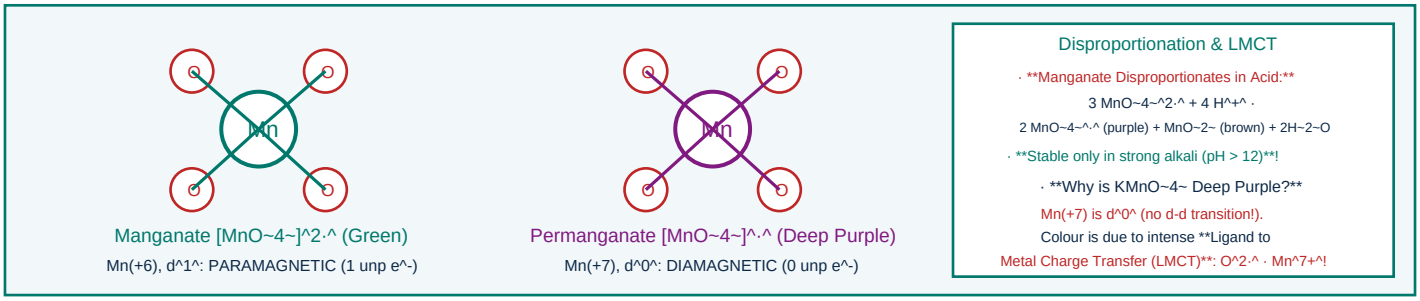


Fig. 13.1 — Tetrahedral structures of manganate (green, paramag.) and permanganate (purple, diamag.).

1. Two-Step Industrial Manufacture

- **Step 1: Oxidative Alkaline Fusion of Pyrolusite:**
- Finely powdered MnO_2 is fused with KOH (or K_2CO_3) in presence of atmospheric air or an oxidizing agent like KNO_3 or $KClO_3$:
- $2 MnO_2 + 4 KOH + O_2 \rightarrow 2 K_2MnO_4 (dark\ green) + 2 H_2O$
- Dark green potassium manganate (K_2MnO_4) is formed.
- **Step 2: Oxidation of Manganate to Permanganate:**
- - **Commercial Electrolytic Route:** Alkaline manganate solution is electrolyzed between iron electrodes. At anode: $MnO_4^{2-} \rightarrow MnO_4^- + e^-$ (oxidation to purple permanganate).
- - **Laboratory Chemical Route (Disproportionation):** Green manganate is acidified or bubbled with CO_2 or Cl_2:
- $3 MnO_4^{2-} (green) + 4 H^+ \rightarrow 2 MnO_4^- (purple) + MnO_2\downarrow (brown) + 2 H_2O$
- $2 K_2MnO_4 + Cl_2 \rightarrow 2 KMnO_4 + 2 KCl$

2. Structural, Magnetic & Optical Comparison

Property	Potassium Manganate (K_2MnO_4)	Potassium Permanganate ($KMnO_4$)
Oxidation State of Mn	+6 ($3d^1$ configuration)	+7 ($3d^0$ configuration)
Geometry & Hybridization	Tetrahedral (d^{3s} or sp^3 with pπ-dπ bonding)	Tetrahedral (d^{3s} or sp^3 with pπ-dπ bonding)
Magnetic Property	**Paramagnetic** (1 unpaired electron, $\mu = 1.73\text{ BM}$)	**Diamagnetic** (0 unpaired electrons)
Colour & Spectroscopy	**Dark Green** (d-d transition + LMCT)	**Intense Deep Purple** (pure LMCT: $O^{2-} \rightarrow Mn^{7+}$)
Solution Stability	Stable ONLY in strongly alkaline solution (pH > 12)	Stable in neutral/dilute acid; decomposes slowly in light

Table 13.1 — Comparative properties of manganate and permanganate.

TRAP
CRITICAL TRAP: $KMnO_4$ has ZERO unpaired d-electrons (d^0), yet it is intensely purple! Never claim its color is due to d-d transition; it is strictly caused by Ligand-to-Metal Charge Transfer (LMCT)!

Volumetric Equivalent Weights Across Three Media (NCERT §8.4)

The oxidizing power and equivalent weight of $KMnO_4$ vary dramatically depending upon the pH of the reaction medium, transferring 5, 3, or 1 electron.

1. The Three Half-Reactions & Equivalent Weights

- **1. Acidic Medium (dilute H_2SO_4):** Mn(+7) is reduced by **5 electrons** to Mn^{2+} (+2):
- $MnO_4^- + 8\ H^+ + 5\ e^- \rightarrow Mn^{2+}\ (faint\ pink) + 4\ H_2O \quad (E^\circ = +1.51\ V)$
- $\mathbf{\text{Equivalent Weight}} = \frac{\text{Molar Mass}}{5} = \frac{158.04}{5} = \mathbf{31.61\ g/eq}$.
- **2. Neutral or Faintly Alkaline Medium (Baeyer's Condition):** Mn(+7) is reduced by **3 electrons** to brown insoluble MnO_2 (+4):
- $MnO_4^- + 2\ H_2O + 3\ e^- \rightarrow MnO_2\downarrow (brown) + 4\ OH^- \quad (E^\circ = +1.23\ V)$
- $\mathbf{\text{Equivalent Weight}} = \frac{\text{Molar Mass}}{3} = \frac{158.04}{3} = \mathbf{52.68\ g/eq}$.
- **3. Strongly Alkaline Medium:** Mn(+7) is reduced by **1 electron** to green manganate MnO_4^{2-} (+6):
- $MnO_4^- + e^- \rightarrow MnO_4^{2-}\ (green) \quad (E^\circ = +0.56\ V)$
- $\mathbf{\text{Equivalent Weight}} = \frac{\text{Molar Mass}}{1} = \mathbf{158.04\ g/eq}$.

2. Quantitative Acidic Redox Reactions

Reducing Agent	Oxidized Product	Ionic Equation	Special Feature
Oxalate ($C_2O_4^{2-}$)	CO_2 gas	$2\ MnO_4^- + 16\ H^+ + 5\ C_2O_4^{2-} \rightarrow 2\ Mn^{2+} + 10\ CO_2\uparrow + 8\ H_2O$	Autocatalyzed by product Mn^{2+} ions!
Fe^{2+} (Mohr's Salt)	Fe^{3+} (Ferric)	$MnO_4^- + 8\ H^+ + 5\ Fe^{2+} \rightarrow Mn^{2+} + 5\ Fe^{3+} + 4\ H_2O$	$KMnO_4$ acts as self-indicator
I^- (Iodide)	I_2 (Iodine)	$2\ MnO_4^- + 16\ H^+ + 10\ I^- \rightarrow 2\ Mn^{2+} + 5\ I_2 + 8\ H_2O$	Forms brown iodine solution
H_2S	S (Colloidal sulfur)	$2\ MnO_4^- + 6\ H^+ + 5\ H_2S \rightarrow 2\ Mn^{2+} + 5\ S\downarrow + 8\ H_2O$	Turbid white colloidal sulfur
NO_2^- (Nitrite)	NO_3^- (Nitrate)	$2\ MnO_4^- + 6\ H^+ + 5\ NO_2^- \rightarrow 2\ Mn^{2+} + 5\ NO_3^- + 3\ H_2O$	Colorless nitrate solution

Table 14.1 — Standard acidic redox titrations using $KMnO_4$.

3. Neutral / Faintly Alkaline Traps (JEE Advanced Core)

- **Oxidation of Iodide to IODATE (Crucial Contrast!):**
- In acidic medium, I^- gives I_2. In neutral/faintly alkaline medium, I^- is oxidized to **Iodate (IO_3^-)**:
- $2\ MnO_4^- + H_2O + I^- \rightarrow 2\ MnO_2\downarrow + IO_3^- + 2\ OH^-$
- **Oxidation of Thiosulfate to SULFATE:**
- $8\ MnO_4^- + 3\ S_2O_3^{2-} + H_2O \rightarrow 8\ MnO_2\downarrow + 6\ SO_4^{2-} + 2\ OH^-$
- **Why HCl cannot be used to acidify $KMnO_4$?** $KMnO_4$ ($E^\circ = +1.51\ V$) oxidizes HCl ($E^\circ = +1.36\ V$) to toxic Cl_2 gas: $2\ MnO_4^- + 16\ HCl \rightarrow 2\ Mn^{2+} + 5\ Cl_2\uparrow + 8\ H_2O$. Dilute H_2SO_4 must be used!

TRAP
SELF-INDICATOR ACTION: $KMnO_4$ acts as its own indicator in titrations. A single excess drop imparts a permanent, unmistakable pale pink color to the solution at the equivalence point!

Inner Transition Series (4f Orbital Filling) (NCERT §8.5)

The 14 lanthanoids span from **Cerium (Ce 58) to Lutetium (Lu 71)** following Lanthanum (La 57). They are characterized by the progressive filling of 4f inner orbitals shielded by outer 5s, 5p, and 6s subshells.

1. Electronic Configurations & Dominant Oxidation States

- **General Valence Configuration: [Xe] $4f^{1-14}$ $5d^{0-1}$ $6s^2$.**
- - 5d orbital has 1 electron only in **La ([Xe] $5d^1$ $6s^2$), Ce ([Xe] $4f^1$ $5d^1$ $6s^2$), Gd ([Xe] $4f^7$ $5d^1$ $6s^2$), and Lu ([Xe] $4f^{14}$ $5d^1$ $6s^2$)**.
- - In all other lanthanoids, 5d electron is shifted into 4f subshell.
- **The Supreme Dominance of +3 State:** All lanthanoids exhibit **+3** as their principal, most stable oxidation state in aqueous solution and solid compounds.
- **Exceptions (+2 and +4 States):**
- - **Ce^{4+} (+4):** Attains empty noble gas **$4f^0$** configuration. Exceptionally powerful oxidant ($E^{\circ} = +1.74\text{ V}$), used in volumetric cerimetry.
- - **Tb^{4+} (+4):** Attains stable half-filled **$4f^7$** configuration. Strong oxidant.
- - **Eu^{2+} (+2):** Attains stable half-filled **$4f^7$** configuration ([Xe] $4f^7$). Strong reducing agent ($E^{\circ}(Eu^{3+}/Eu^{2+}) = -0.43\text{ V}$).
- - **Yb^{2+} (+2):** Attains completely filled **$4f^{14}$** configuration. Strong reducing agent.

2. Lanthanoid Contraction: Causes & Consequences

Steady decrease in atomic and ionic radii from Ce^{3+} (103 pm) to Lu^{3+} (86 pm) due to poor shielding of diffuse 4f electrons.

Consequence	Underlying Chemical / Physical Cause	Practical Observation / Impact
Basic Strength of Hydroxides	Smaller ionic size increases covalent character (Fajans)	**$Ce(OH)_3$ is most basic, $Lu(OH)_3$ is least basic**
Separation Difficulty	Nearly identical chemical properties and ionic sizes	Requires fractional crystallization or ion-exchange chromatography
4d/5d Twin Pairs	Cancels normal group radius expansion	Zr/Hf, Nb/Ta, Mo/W have identical sizes and co-occur in nature

Table 15.1 — Major consequences of Lanthanoid Contraction.

3. Chemical Reactivity, Sharp Spectra & Mischmetal

- **Reactivity:** Electropositive metals, resembling Calcium: react with water yielding H_2, burn in O_2 forming Ln_2O_3, react with halogens forming LnX_3.
- **Sharp f–f Absorption Bands:** 4f electrons are deeply buried inside the atom and shielded by $5s^2$ $5p^6$ electrons → crystal fields cannot broaden orbital levels → absorption spectra consist of **extremely sharp lines** (used in Nd:YAG laser rods).
- **Mischmetal:** Alloy consisting of 95% Lanthanoid metals (Ce 50%, La 40%), 5% Fe, and traces of S, C, Ca. Used in cigarette lighter flints and tracer bullets.

TRAP
RADIOACTIVITY CHECK: Among all 14 lanthanoids, **ONLY Promethium (Pm 61)** is radioactive and synthetic; all other 13 lanthanoids are non-radioactive stable elements!

Inner Transition Series (5f Orbital Filling) (NCERT §8.6)

The 14 actinoids span from **Thorium (Th 90) to Lawrencium (Lr 103)** following Actinium (Ac 89). They involve the progressive population of 5f subshell.

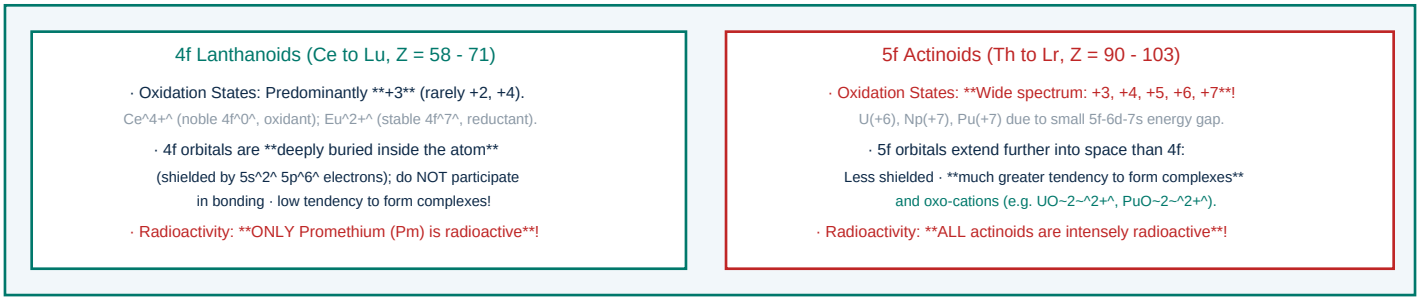


Fig. 16.1 — Comparative properties: 4f Lanthanoids vs 5f Actinoids.

1. Electronic Configuration & Multiple Oxidation States

- **General Valence Configuration: $[Rn]\ 5f^{1-14}\ 6d^{0-2}\ 7s^2$.**
- **Wide Spectrum of Oxidation States:** Unlike lanthanoids (dominated by +3), early actinoids exhibit multiple oxidation states: **Th (+4), Pa (+5), U (+3, +4, +5, +6), Np (+3, +4, +5, +6, +7), Pu (+3, +4, +5, +6, +7)**.
- **Physical Reason:** In actinoids, **5f, 6d, and 7s energy levels are extremely close in energy** (very small energy gap), allowing all outer electrons to participate in chemical bonding!

2. Actinoid Contraction vs Lanthanoid Contraction

- **Actinoid Contraction:** Progressive decrease in size of atoms and M^{3+}/M^{4+} ions across the series.
- **Comparison: Actinoid contraction from element to element is GREATER than lanthanoid contraction!**
- **Physical Cause:** 5f orbitals extend further from the nucleus than 4f orbitals, resulting in **even poorer shielding** of nuclear charge. Consequently, effective nuclear charge (Z_{eff}) increases more steeply across the actinoid series.

3. Rigorous Comparison: Lanthanoids vs Actinoids

Feature	Lanthanoids (4f Series)	Actinoids (5f Series)
Oxidation States	Predominantly **+3** (rarely +2, +4)	**Multiple states: +3, +4, +5, +6, +7**
Energy Gap	Large gap between 4f, 5d, and 6s orbitals	Very small gap between 5f, 6d, and 7s orbitals
Complexing Ability	Low tendency (4f orbitals deeply buried)	**High tendency** (5f orbitals extend spatially)
Oxocations	Do NOT form oxocations	Readily form stable oxocations: **$[UO_2]^{2+}$, $[PuO_2]^{2+}$**
Basic Strength	Hydroxides are less basic	Hydroxides are more basic
Radioactivity	Only Promethium (Pm) is radioactive	**ALL actinoids are intensely radioactive!**

Table 16.1 — Comprehensive Lanthanoid vs Actinoid comparison.

> TRAP
> TRANSURANIC ELEMENTS: Elements beyond Uranium (Z > 92, from Neptunium onwards) are completely synthetic, man-made transuranic elements produced in nuclear reactors!

Industrial Compounds & Analytical Chemistry of Ag, Cu, and Zn

The chemistry of late 3d and 4d metals involves distinctive precipitation, complexation, and redox reactions central to qualitative analysis and industrial technology.

1. Silver Chemistry: AgNO3 & Silver Halides

- **Silver Nitrate ($AgNO_3$, Lunar Caustic):** *Prepared by dissolving pure Ag in dilute HNO_3:* 3 Ag + 4 HNO_3 \rightarrow 3 AgNO_3 + NO\uparrow + 2 H_2O.
- - **Thermal Decomposition:** 2 AgNO_3 \xrightarrow{\Delta} 2 Ag + 2 NO_2\uparrow + O_2\uparrow *(decomposes directly to metallic silver).*
- - **Caustic Stain:** *On skin/organic matter, produces a permanent black metallic silver stain.*
- **Silver Halides Ammonia & Hypo Solubility:**
- - **AgCl (White curdy ppt):** *Soluble in dilute aqueous NH_3:* AgCl + 2 NH_3 \rightarrow [Ag(NH_3)_2]^+ + Cl^-.
- - **AgBr (Pale yellow ppt):** *Sparingly soluble in dilute NH_3, soluble in concentrated NH_3; dissolves in hypo:* AgBr + 2 Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr *(photographic fixing).*
- - **AgI (Bright yellow ppt):** *Insoluble in both dilute and concentrated NH_3; soluble only in KCN or hypo.*

2. Copper Chemistry: Blue Vitriol & Cuprous Compounds

- **Copper Sulfate Pentahydrate ($CuSO_4 \cdot 5H_2O$, Blue Vitriol):**
- - **Structural Insight:** *Four water molecules coordinate to Cu^{2+} in a square plane ($[Cu(H_2O)_4]^{2+}$), while the fifth water molecule is held by* **hydrogen bonds** *between coordinated waters and sulfate:* [Cu(H_2O)_4]SO_4\cdot H_2O*!*
- - **Thermal Dehydration:**
- $CuSO_4\cdot 5H_2O \xrightarrow{100^\circ C} CuSO_4\cdot H_2O \xrightarrow{250^\circ C} CuSO_4 (white) \xrightarrow{750^\circ C} CuO + SO_3$
- **Reaction with KI (Iodometric Titration Foundation):**
- $2 CuSO_4 + 4 KI \rightarrow 2 CuI\downarrow (white) + I_2 + 2 K_2SO_4$
- *Liberated iodine is titrated against standard hypo solution using starch indicator.*

3. Zinc Chemistry: Amphoterism & Lithopone

- **Amphoteric Nature:** *Zn, ZnO, and $Zn(OH)_2$ react with both acids and strong bases:*
- $Zn + 2 NaOH \rightarrow Na_2ZnO_2 (sodium\ zincate) + H_2\uparrow$
- $ZnO + 2 NaOH + H_2O \rightarrow Na_2[Zn(OH)_4]$
- **Zinc Vitriol ($ZnSO_4 \cdot 7H_2O$, White Vitriol):** *Isomorphous with Epsom salt ($MgSO_4 \cdot 7H_2O$).*
- **Lithopone:** *Equimolar mixture of* **$BaSO_4$ + ZnS**, *prepared by precipitation:* BaS + ZnSO_4 \rightarrow BaSO_4\downarrow + ZnS\downarrow. *Non-poisonous white paint pigment (does not blacken in H_2S atmosphere!).*

> TRAP
> *EXAM TRAP: $CuCl_2$ is green in concentrated solution, turns sky-blue upon dilution due to transformation of $[CuCl_4]^{2-}$ (yellow/green) into $[Cu(H_2O)_6]^{2+}$ (blue)!*

Iron Complexes, Brown Ring Test & High-Yield Exam Traps

Iron exhibits intricate coordination and redox chemistry, providing foundation tests in qualitative inorganic analysis.

1. Mohr's Salt, Green Vitriol & Prussian Blue

- **Mohr's Salt (Ferrous Ammonium Sulfate): $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$.**
- *- Primary volumetric standard for Fe^{2+} because it is **resistant to atmospheric oxidation** in solid form (unlike $FeSO_4 \cdot 7H_2O$ which rapidly oxidizes to basic ferric sulfate).*
- **Green Vitriol ($FeSO_4 \cdot 7H_2O$):**
- *- Thermal Decomposition:* $2\ FeSO_4 \cdot 7H_2O \xrightarrow{\Delta} Fe_2O_3$ (red-brown) $+ SO_2\uparrow + SO_3\uparrow + 14\ H_2O$.
- **Prussian Blue & Turnbull's Blue Identity:**
- *- Fe^{3+} + potassium ferrocyanide* $\rightarrow$ ***Prussian Blue***: $Fe^{\{3+\}} + [Fe(CN)_6]\{4-\} \rightarrow KFe^{\{III\}[Fe}\{II\}(CN)_6]$.
- *- Fe^{2+} + potassium ferricyanide* $\rightarrow$ ***Turnbull's Blue***: $Fe^{\{2+\}} + [Fe(CN)_6]\{3-\} \rightarrow KFe^{\{III\}[Fe}\{II\}(CN)_6]$.
- *- **Modern Spectroscopic Truth:** Both are chemically IDENTICAL: **iron(III) hexacyanoferrate(II)**! The intense deep blue color arises from **inter-valence charge transfer (IVCT)** between Fe^{2+} and Fe^{3+}.*

2. The Brown Ring Nitrate Complex

*Formed at the interface in the brown ring test for nitrate: · **Formula: $[Fe(H_2O)_5(NO)]SO_4$.** · **Electronic Nature:** Electron transfer from NO to Fe^{3+} establishes an equilibrium between $Fe(+1)$-NO^+ and $Fe(+3)$-NO^- with **3 unpaired electrons** ($\mu_{eff} \approx 3.87\text{ BM}$). · **Colour Origin:** Intense brown color is due to **charge transfer** between iron and nitrosyl ligand.*

3. JEE Advanced Master Reasoning Bank

Exam Question	Fundamental Chemical Principle	Precise Exam Justification
Why is Sc^{3+} colourless while Ti^{3+} is purple?	d-d electronic excitation	Sc^{3+} is [Ar] $3d^0$ (no d-electrons). Ti^{3+} is $3d^1$; visible photon excites electron from t_{2g} to e_g.
Why is atomization enthalpy of Zn lowest in 3d?	Metallic lattice cohesion	Zn has completely filled $3d^{10}\ 4s^2$; no unpaired d-electrons participate in covalent d-d overlap.
Why is Cr^{2+} reducing while Mn^{3+} is oxidizing?	Electronic shell stability	Cr^{2+} oxidizes to Cr^{3+} (stable half-filled t_{2g}^3 CFSE). Mn^{3+} reduces to Mn^{2+} (stable $3d^5$ exchange energy).
Why does Cu^+ disproportionate in water?	Hydration vs Ionization balance	Cu^{2+} has high charge density, giving massive negative hydration enthalpy ($\Delta_{hyd}H^\circ = -2121$ kJ/mol) overcoming IE_2.

Table 18.1 — Essential JEE Advanced reasoning traps and authoritative explanations.

TRAP

EXAM TRAP: Sodium nitroprusside $Na_2[Fe(CN)_5(NO)]$ reacts with sulfide ion (S^{2-}) to give an intense purple/violet coloration of $[Fe(CN)_5(NOS)]^{4-}$ — classic confirmatory test for sulfide!